Mining GOLD Samples for Conditional GANs

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Introduction

- **Conditional GANs (cGANs)** have gained a considerable attention due to its class-wise controllability and superior quality



* Image from BigGAN (ICLR 2019)

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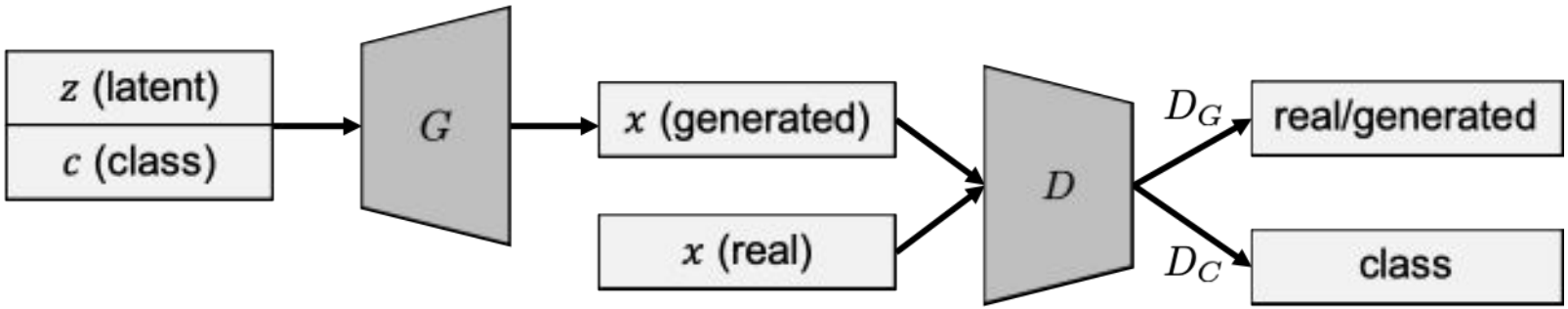
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- We present three applications of the GOLD estimator, which improves the **training**, **inference**, and reduce **labeling cost** of cGANs
- We demonstrate that the **GOLD-based methods** outperform corresponding baselines for all three applications on various datasets

Outline

1. Preliminary: Conditional GANs
2. Gap of Log-densities (GOLD)
3. Applications of the GOLD Estimator
4. Experiments

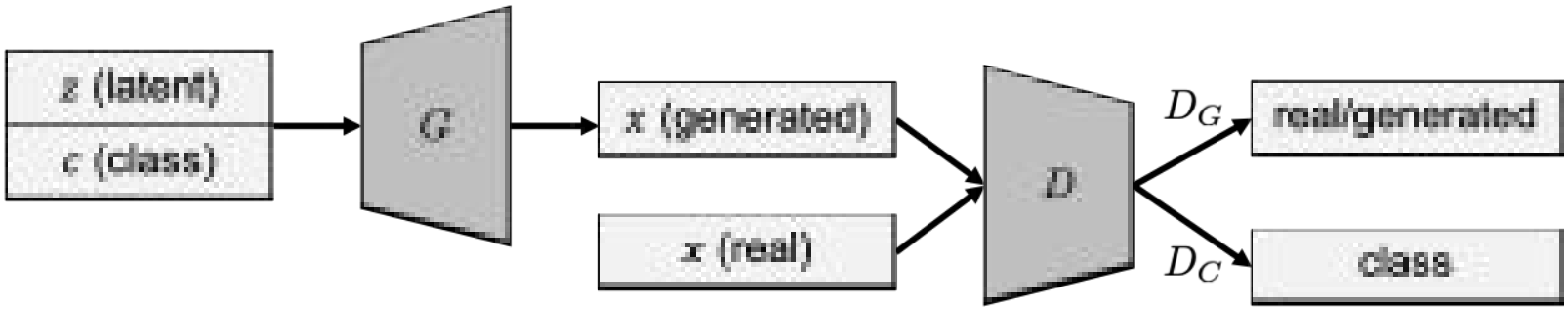
Preliminary: Conditional GANs



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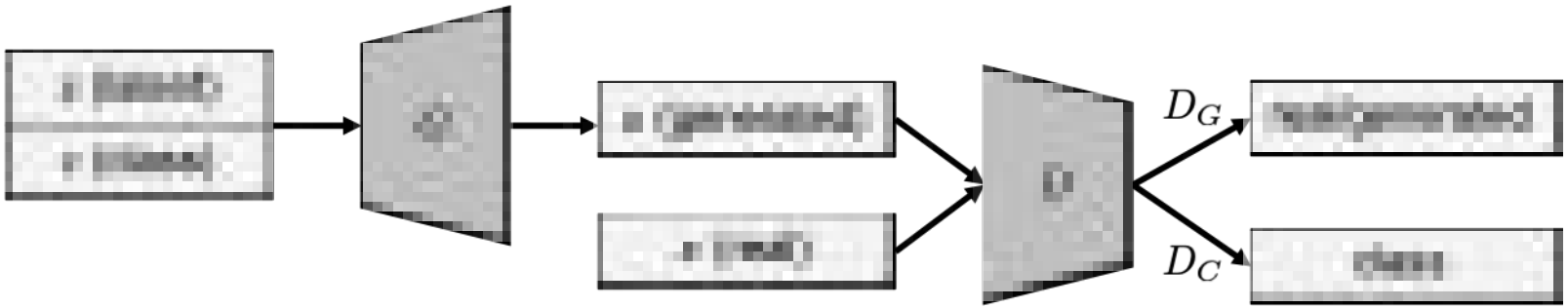


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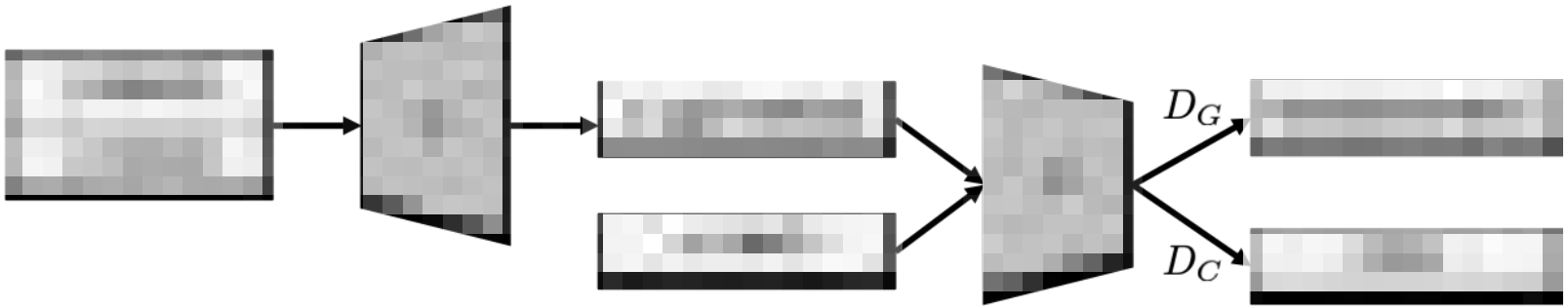


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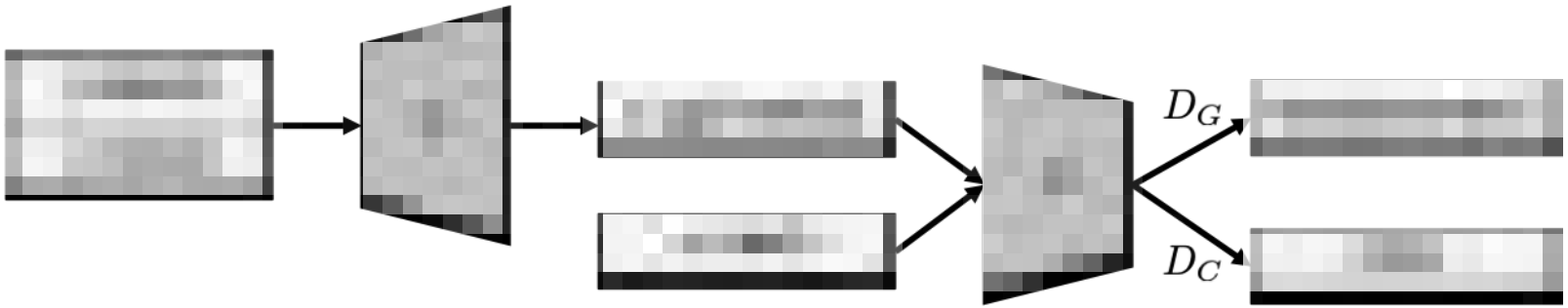


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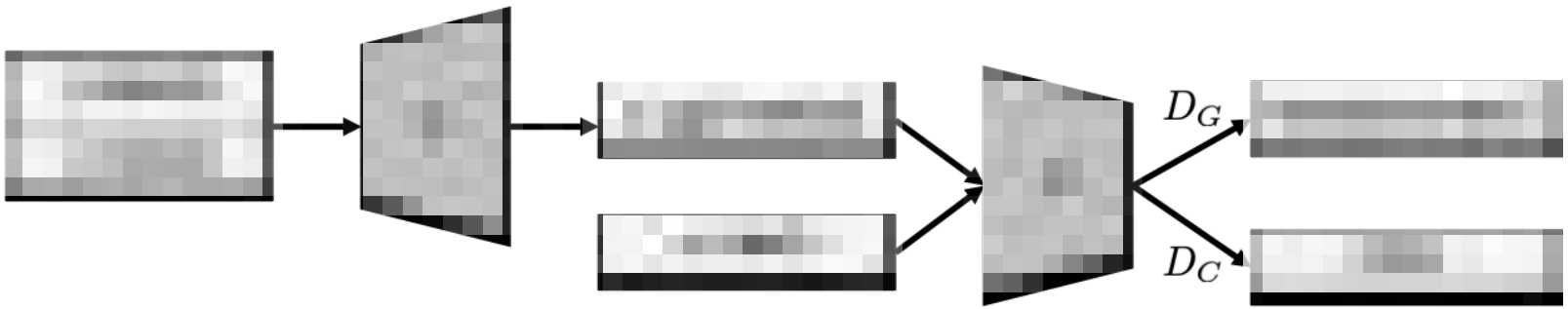


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  - the **classifier** branch $D_C : x \mapsto c$ which learns the conditional $p(c|x)$

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Preliminary: Conditional GANs



ACGAN is trained by both GAN loss $\mathcal{L}_{\text{GAN}}$ and classifier loss $\mathcal{L}_{\text{AC}}$:[1][2]

$$\mathcal{L}_{\text{GAN}} = \mathbb{E}_{p_{\text{data}}(x,c)}[-\log D_G(x)] + \mathbb{E}_{p_g(z,c)}[\log D_G(G(z,c))]$$

$$\mathcal{L}_{\text{AC}} = \mathbb{E}_{p_{\text{data}}(x,c)}[-\log D_C(c|x)] + \lambda_c \mathbb{E}_{p_g(z,c)}[-\log D_C(c|G(z,c))]$$

[1] Generator and discriminator minimizes $-\mathcal{L}_{\text{GAN}} + \mathcal{L}_{\text{AC}}$ and $\mathcal{L}_{\text{GAN}} + \mathcal{L}_{\text{AC}}$, respectively

[2] Original ACGAN simply sets $\lambda_c = 1$, but we find that using small λ_c works better

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Here, the **marginal** and **conditional** terms measure the *generation quality* and *class accuracy* of the generated samples, respectively

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To sum up, the GOLD estimator is given by

$$d(x, c_x) := \begin{cases} \log \frac{D_G(x)}{1-D_G(x)} + \log D_C(c_x|x) & \text{if } x \text{ is a } \textit{generated} \text{ sample} \\ \log \frac{D_G(x)}{1-D_G(x)} - \log D_C(c_x|x) & \text{if } x \text{ is a } \textit{real} \text{ sample} \end{cases} \tag{2}$$

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Note that the conditional term of (2) has an **opposite sign** for the generated and real samples, which *matches the sign* to the marginal terms

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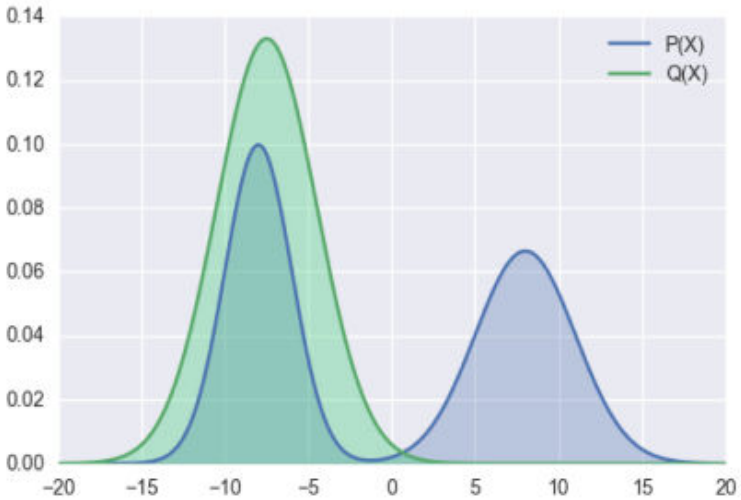
Improving Training of cGANs (Example Re-weighting)

- **High GOLD** values imply that the sample (x, c_x) is **under-estimated** with respect to the joint distribution $p(x, c)$, and vice versa

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* Image from Tutorial Example

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- We propose a re-weighted loss calibrated by the GOLD estimator:[1][2]

$$\mathcal{L}'_{\text{GAN}} = \mathbb{E}_{p_{\text{data}}(x,c)}[-\log D_G(x)] + \mathbb{E}_{p_g(z,c)}[d(G(z,c),c)^\beta \log D_G(G(z,c))]$$
$$\mathcal{L}'_{\text{AC}} = \mathbb{E}_{p_{\text{data}}(x,c)}[-\log D_C(c|x)] + \lambda_c \mathbb{E}_{p_g(z,c)}[-d(G(z,c),c)^\beta \log D_C(c|G(z,c))]$$

[1] $\beta \geq 0$ is a hyper-parameter to control the level re-weighting and we use $x^\beta = -|x|^\beta$ for $x < 0$. We simply choose $\beta = 1$ for discriminator loss and $\beta = 0$ for generator loss

[2] We apply re-weighting scheme after sufficiently training the model with original loss

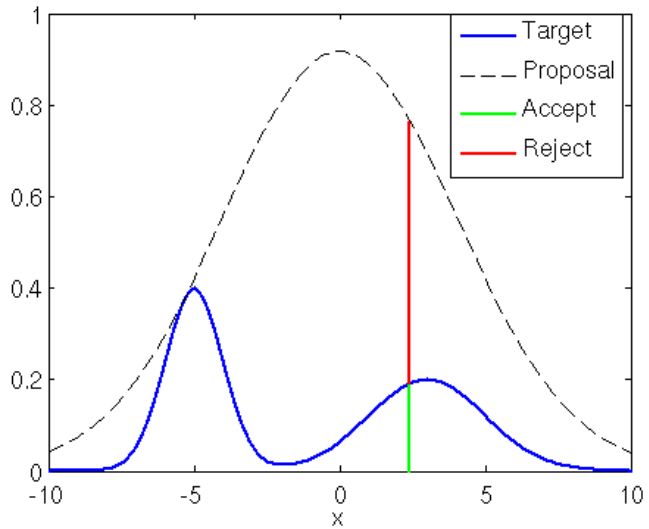
Improving Inference of cGANs (Rejection Sampling)

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* Image from The Clever Machine

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- **Rejection sampling:** Using $p_g(z, c)$ as a proposal distribution, we *accept* it with probability $\frac{p_{\text{data}}(z,c)}{M p_g(z,c)}$ and *reject* otherwise[1][2]
- The acceptance rate can be approximated by the GOLD estimator:

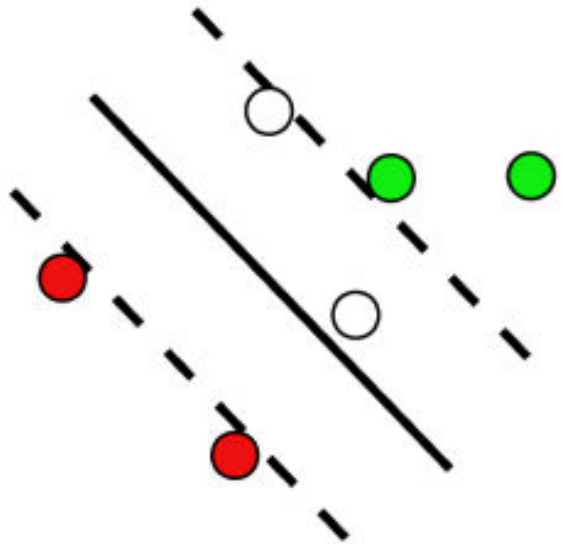
$$r(x) := \frac{1}{M} \exp\left(d(x, c_x)\right) = \frac{1}{M} \exp\left(\log \frac{D_G(x)}{1 - D_G(x)} + \log D_C(c_x|x)\right)$$

[1] M is set to be the maximum of $\exp(d(x, c_x))$ among samples

[2] This technique is proposed in Azadi et al. Discriminator Rejection Sampling (DRS). ICLR 2019. We extend the algorithm to the conditional GANs.

Reducing Labeling Costs of cGANs (Active Learning)

- **Active learning:** From the *pool* of unlabeled samples, choose the samples to ask *labels* which improve the model mostly



* Image from George Nichols's slide

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- It is natural to ask **high GOLD** (under-estimated) samples, but c_x is unknown! Hence, we take an expectation over the probability of D_C[1]

$$-\log D_C(c_x|x) \approx \mathbb{E}_{c\sim D_C(c|x)}[-\log D_C(c|x))] = \mathcal{H}[D_C(c|x)]$$

[1] $\mathcal{H}$ is an entropy function

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- The GOLD estimator for the unlabeled real samples are given by[2]

$$d_{\texttt{unlabel}}(x) := \log \frac{D_G(x)}{1-D_G(x)} + \mathcal{H}[D_C(c|x)]$$

[1] $\mathcal{H}$ is an entropy function

[2] It is a combination of *denstiy*-based and *uncertainty*-based methods for the active learning for classifiers. We pick the samples with largest $d_{\texttt{unlabel}}(x)$ values

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Experiment Setting

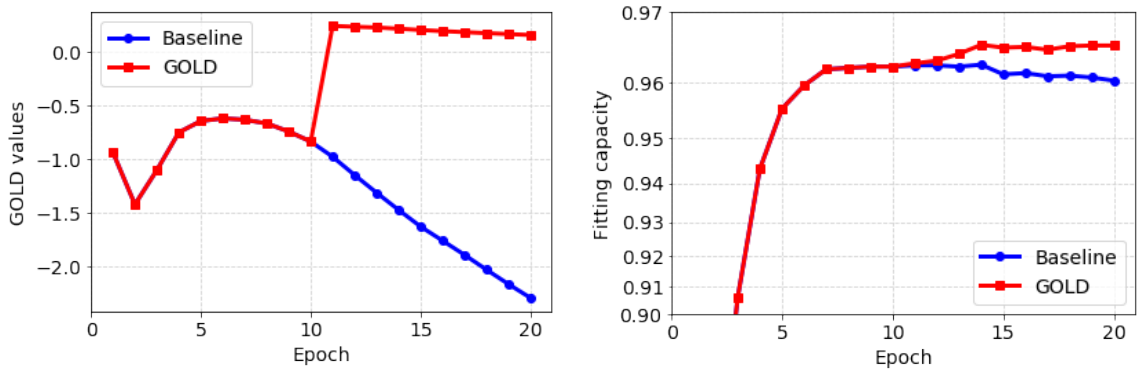
- We demonstrate the effectiveness of the GOLD estimator for three applications under **1** synthetic and **8** image datasets
- We evaluate the fitting capacity[1][2], which measures the accuracy of **real** samples under a classifier trained with **generated** samples
- We average **5** trials for image datasets and **25** trials for synthetic dataset. See the paper for the detailed experiment settings

[1] Lesort et al. Training Discriminative Models to Evaluate Generative Ones. arXiv 2018.

[2] It is also called as classification accuracy score (CAS). Ravuri and Vinyals. Classification Accuracy Score for Conditional Generative Models. NeurIPS 2019.

Example Re-weighting (Training) Experiments

- Ours stabilizes the GOLD estimator and improves the training

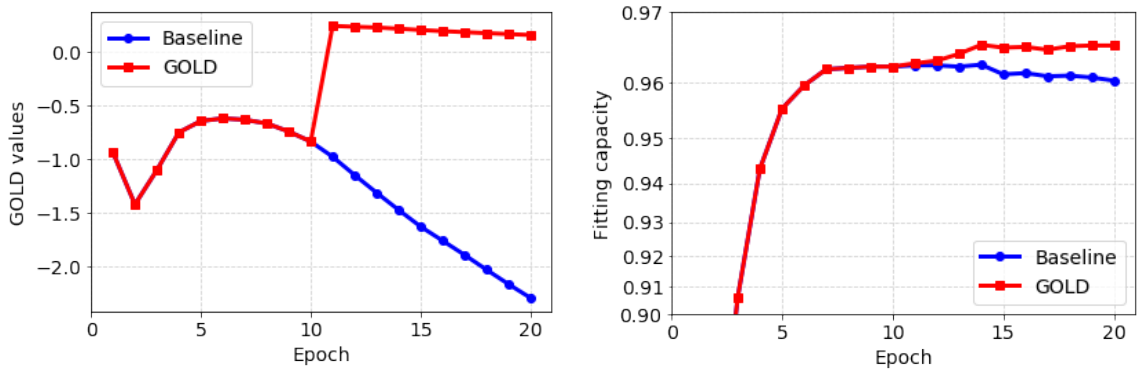


(a) GOLD estimator (b) Fitting capacity

Figure 1: Training curve of the mean of GOLD estimator and the fitting capacity

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Table 1: Fitting capacity for example re-weighting under various datasets

	MNIST	FMNIST	SVHN	CIFAR-10	STL-10	LSUN
Baseline	96.43±0.17	77.97±1.24	74.43±0.71	36.76±0.99	36.73±0.64	26.35±0.82
GOLD	**96.62**±0.15	**78.34**±1.11	**76.71**±0.94	**37.06**±1.38	**37.65**±0.71	**28.21**±0.86

Example Re-weighting (Training) Experiments

- Ours outperforms for various supervisions and larger-scale datasets

Table 2: Fitting capacity for example re-weighting under various supervisions

	Dataset	1%	5%	10%	20%	50%	100%
Baseline	SVHN	72.41±1.30	72.99±1.65	73.15±0.96	73.18±1.28	74.04±1.26	74.33±0.71
GOLD		**75.01±1.93**	**75.58±0.86**	**75.78±0.74**	**76.04±1.93**	**76.25±1.40**	**76.71±0.94**
Baseline	CIFAR-10	17.99±0.78	18.42±0.71	21.84±1.14	23.13±1.95	35.41±1.03	36.76±0.99
GOLD		**18.28±0.65**	**19.15±0.97**	**21.91±2.56**	**23.89±2.02**	34.95±1.11	**37.06±1.38**

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GOLD		**75.01**±1.93	**75.58**±0.86	**75.78**±0.74	**76.04**±1.93	**76.25**±1.40	**76.71**±0.94
Baseline	CIFAR-10	17.99±0.78	18.42±0.71	21.84±1.14	23.13±1.95	35.41±1.03	36.76±0.99
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Table 3: Fitting capacity (%) for example re-weighting under larger-scale datasets

	Imagenette	Imagewoof
Baseline	20.84±3.39	18.60±1.89
GOLD	**23.64**±3.60	**19.76**±0.90

Rejection Sampling (Inference) Experiments

- Ours improves the sample quality by controlling rejection rate p

Table 4: Fitting capacity for rejection sampling under various datasets

	MNIST	FMNIST	SVHN	CIFAR-10	STL-10	LSUN
Baseline	96.05±0.41	77.94±0.83	73.58±0.72	35.15±0.51	34.33±0.30	26.43±0.14
GOLD	**96.17**±0.63	**78.25**±0.30	**75.06**±0.71	**35.98**±1.15	**35.21**±1.02	**26.79**±0.42

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Table 5: Fitting capacity for rejection sampling under CIFAR-10 and various p

Baseline	p = 0.1	p = 0.3	p = 0.5	p = 0.7	p = 0.9
35.15±0.51	35.80±0.42	35.87±0.61	**35.98±1.15**	35.85±0.53	35.33±0.53

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	MNIST	FMNIST	SVHN	CIFAR-10	STL-10	LSUN
Baseline	96.05±0.41	77.94±0.83	73.58±0.72	35.15±0.51	34.33±0.30	26.43±0.14
GOLD	**96.17**±0.63	**78.25**±0.30	**75.06**±0.71	**35.98**±1.15	**35.21**±1.02	**26.79**±0.42

Table 5: Fitting capacity for rejection sampling under CIFAR-10 and various p

Baseline	p = 0.1	p = 0.3	p = 0.5	p = 0.7	p = 0.9
35.15±0.51	35.80±0.42	35.87±0.61	**35.98**±1.15	35.85±0.53	35.33±0.53

Table 6: Fitting capacity for rejection sampling under larger-scale datasets

	Imagenette	Imagewoof
Baseline	16.89±3.51	16.44±1.97
GOLD	**17.21**±2.08	**16.71**±2.46

Rejection Sampling (Inference) Experiments

- Ours produce diverse samples with good quality

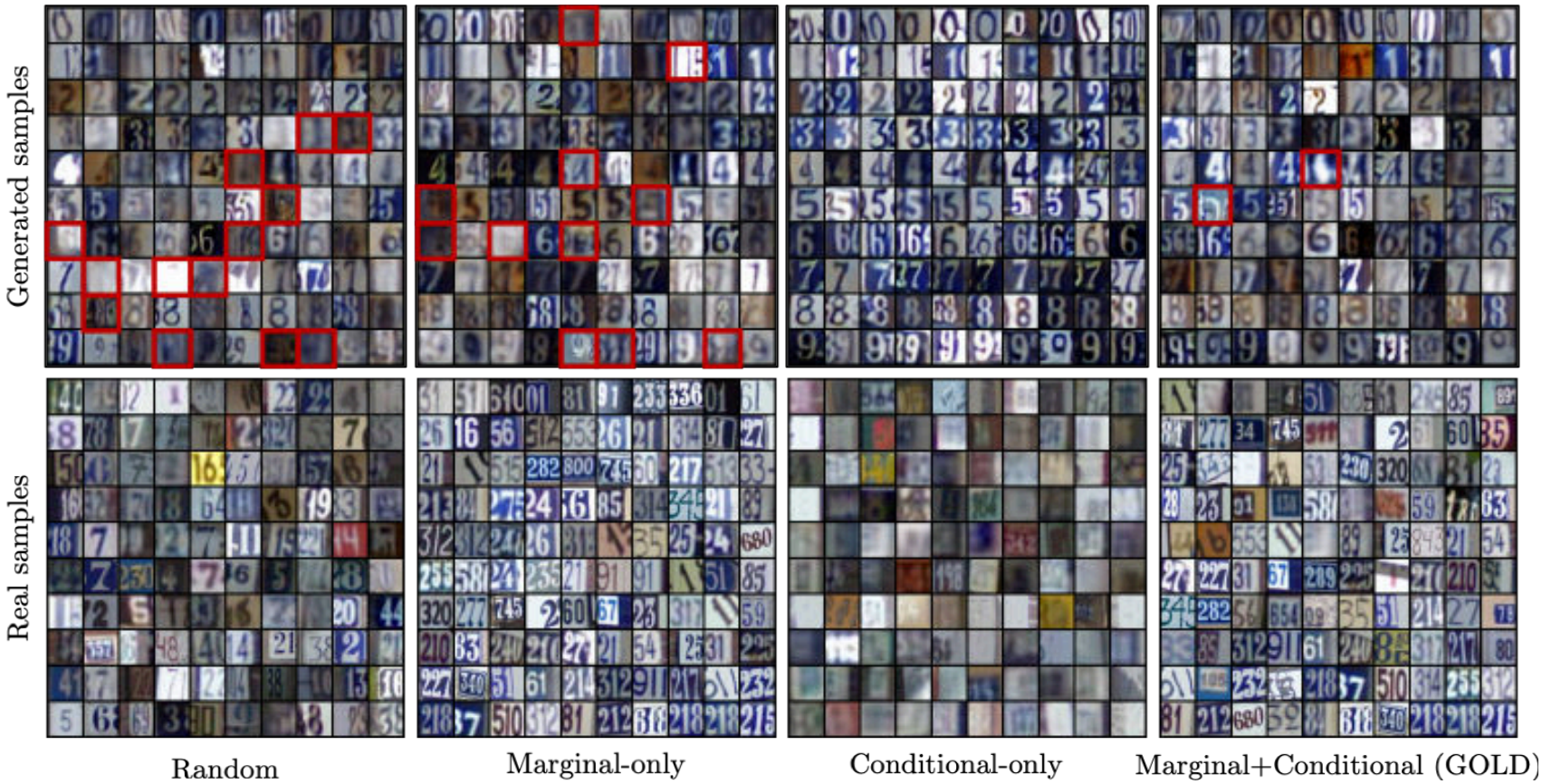


Figure 2: Generated/real samples with high marginal, conditional, and combined (GOLD) values. Red box indicates the uncertain and/or wrong classes

Active Learning (Labeling Cost) Experiments

- Ours select under-estimated samples, and quickly learn distributions

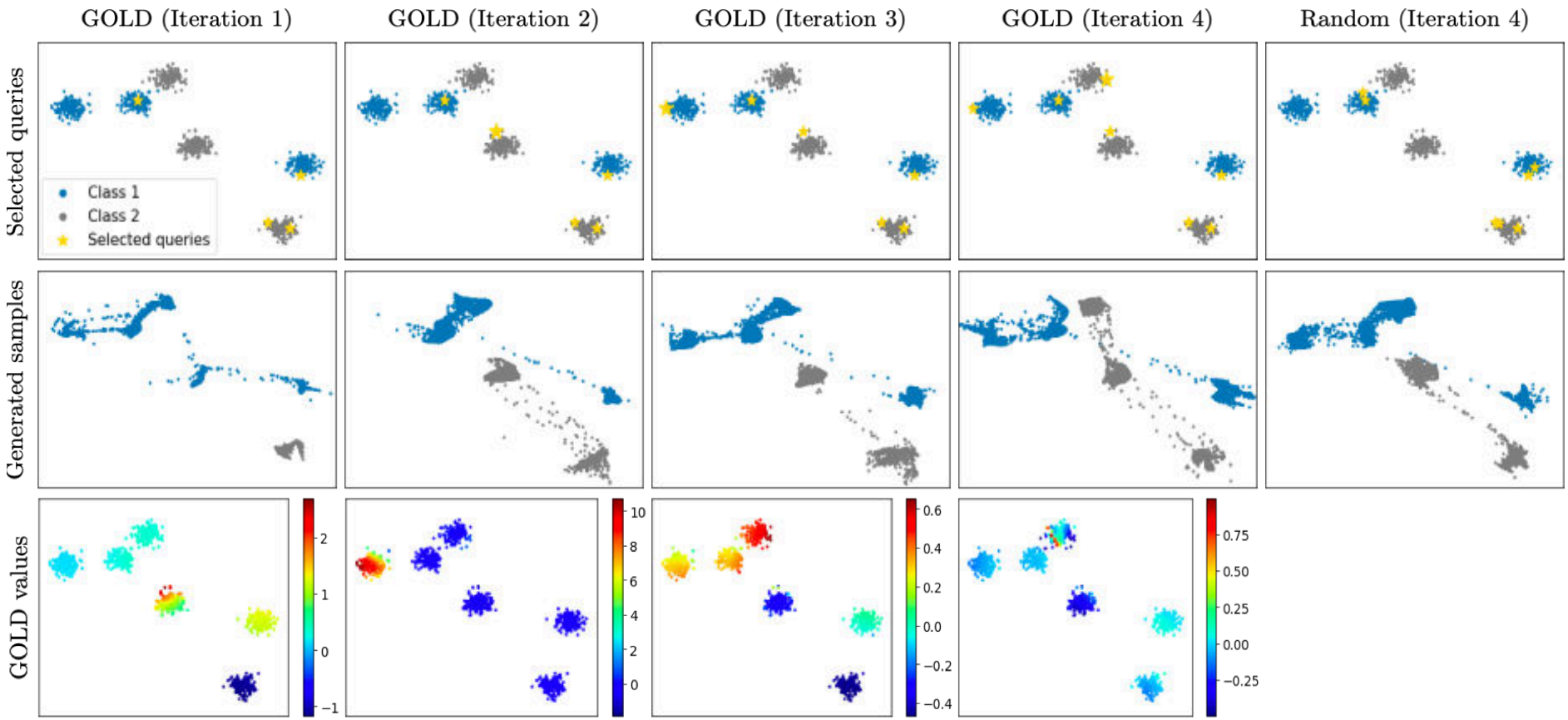
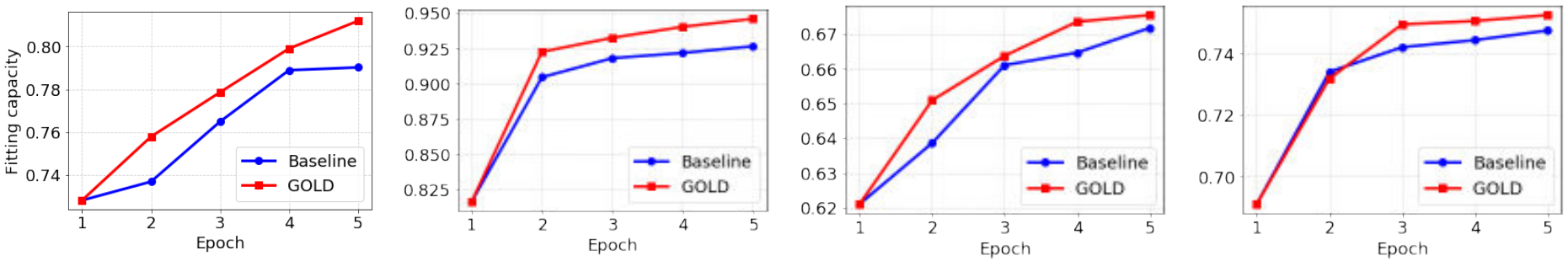


Figure 3: Visualization of the query selection based on the GOLD estimator

Active Learning (Labeling Cost) Experiments

- Ours select under-estimated samples, and quickly learn distributions



(a) Synthetic (b) MNIST (c) FMNIST (d) SVHN

Figure 4: Fitting capacity for active learning under various datasets

Discussion and Conclusion

- We propose the GOLD estimator, a simple yet effective **quality measure** for the real/generated samples of cGANs
- We also present three applications of the GOLD estimator:
  - Improving **training** of cGANs via example re-weighting
  - Improving **inference** of cGANs via rejection sampling
  - Reducing **labeling cost** of cGANs via active learning
- Investigating other applications, e.g., **outlier detection** or training under **noisy labels** would also be an interesting future direction